2.17 Trigonometry: Double Angle Formulae

Worked Solutions

1)

$$\begin{aligned}\sin 2x &\equiv \sin(x+x) \equiv \sin x \cos x + \cos x \sin x \\ &\equiv 2\sin x \cos x\end{aligned}$$

2)

$$\begin{aligned}\cos 2x &\equiv \cos(x+x) \equiv \cos x \cos x - \sin x \sin x \\ &\equiv \cos^2 x - \sin^2 x\end{aligned}$$

3)

$$\begin{aligned}\cos 2x &\equiv \cos^2 x - \sin^2 x \equiv \cos^2 x - (1 - \cos^2 x) \\ &\equiv 2\cos^2 x - 1\end{aligned}$$

4)

$$\begin{aligned}\cos 2x &\equiv \cos^2 x - \sin^2 x \equiv (1 - \sin^2 x) - \sin^2 x \\ &\equiv 1 - 2\sin^2 x\end{aligned}$$

5)

$$\tan 2x \equiv \tan(x+x) \equiv \frac{\tan x + \tan x}{1 - \tan x \tan x} \equiv \frac{2\tan x}{1 - \tan^2 x}$$

6)

$$\begin{aligned}\tan 2x &\equiv \frac{\sin 2x}{\cos 2x} \equiv \frac{2\sin x \cos x}{\cos^2 x - \sin^2 x} \equiv \frac{\dfrac{2\sin x \cos x}{\cos^2 x}}{\dfrac{\cos^2 x - \sin^2 x}{\cos^2 x}} \\ &\equiv \frac{\dfrac{2\sin x}{\cos x}}{1 - \dfrac{\sin^2 x}{\cos^2 x}} \equiv \frac{2\tan x}{1 - \tan^2 x}\end{aligned}$$

7 a)

$$2\sin 2x \cos 2x \equiv \sin(2 \times 2x) \equiv \sin 4x$$

7 b)

$$2\cos^2 3x - 1 \equiv \cos(2 \times 3x) \equiv \cos 6x$$

7 c)

$$1 - \sin^2 4x \equiv \cos(2 \times 4x) \equiv \cos 8x$$

7 d)

$$\begin{aligned} 4\sin \tfrac{1}{2}x \cos \tfrac{1}{2}x &\equiv 2(2\sin \tfrac{1}{2}x \cos \tfrac{1}{2}x) \\ &\equiv 2\sin(2 \times \tfrac{1}{2}x) \equiv 2\sin x \end{aligned}$$

7 e)

$$\begin{aligned} 6\cos^2 \tfrac{1}{3}x - 6\sin^2 \tfrac{1}{3}x &\equiv 6(\cos^2 \tfrac{1}{3}x - \sin^2 \tfrac{1}{3}x) \\ &\equiv 6\cos(2 \times \tfrac{1}{3}x) \equiv 6\cos \tfrac{2}{3}x \end{aligned}$$

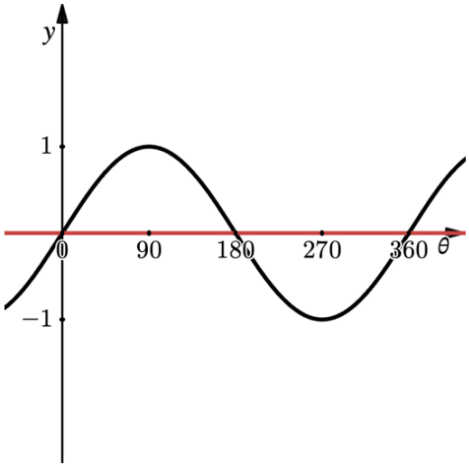
7 f)

$$\begin{aligned} 6\cos^2 \tfrac{3}{2}x - 3 &\equiv 3(2\cos^2 \tfrac{3}{2}x - 1) \\ &\equiv 3\cos(2 \times \tfrac{3}{2}x) \equiv 3\cos 3x \end{aligned}$$

8 a)

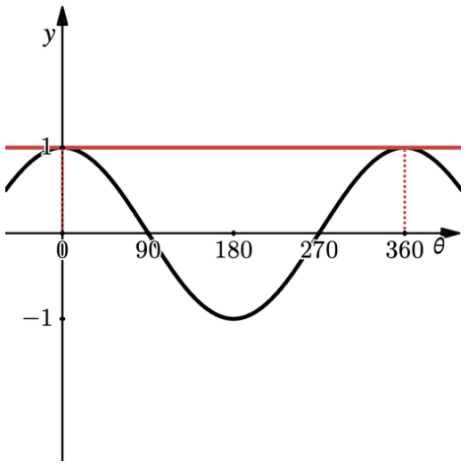
$$\sin 2\theta - 2\sin\theta = 0 \text{ for } 0^\circ \leq \theta < 360^\circ$$
$$2\sin\theta\cos\theta - 2\sin\theta = 0$$
$$\sin\theta\cos\theta - \sin\theta = 0$$
$$\sin\theta(\cos\theta - 1) = 0$$
$$\sin\theta = 0 \text{ or } \cos\theta = 1$$

First solve $\sin\theta = 0$ for $0^\circ \leq \theta < 360^\circ$:



$$\theta = 0^\circ$$
$$\theta = 180^\circ$$

Now solve $\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$:

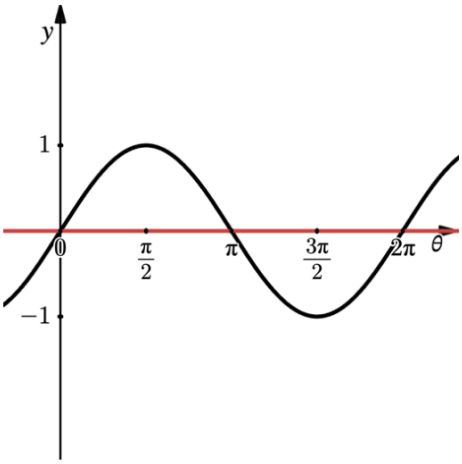


$$\Rightarrow \theta = 0^\circ$$
$$\Rightarrow \theta = \{0^\circ, 180^\circ\}$$

8 b)

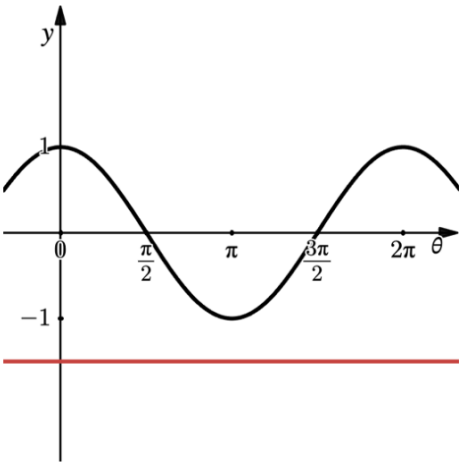
$$\sin 2\theta + 5\sin\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$2\sin\theta\cos\theta + 5\sin\theta = 0$$
$$\sin\theta(2\cos\theta + 5) = 0$$
$$\sin\theta = 0 \text{ or } \cos\theta = -\frac{5}{2}$$

First solve $\sin\theta = 0$ for $0 \leq \theta < 2\pi$:



$$\theta = 0$$
$$\theta = \pi$$

As for $\cos\theta = -\frac{5}{2}$, there are no solutions:

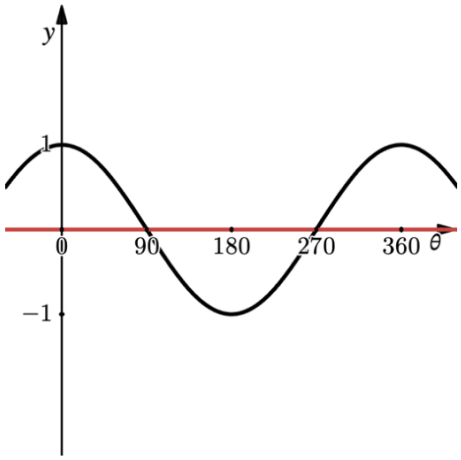


$$\Rightarrow \theta = \{0, \pi\}$$

8 c)

$$3\sin 2\theta = 2\cos\theta \text{ for } 0^\circ \leq \theta < 360^\circ$$
$$3 \times 2\sin\theta\cos\theta - 2\cos\theta = 0$$
$$3\sin\theta\cos\theta - \cos\theta = 0$$
$$\cos\theta(3\sin\theta - 1) = 0$$
$$\cos\theta = 0 \text{ or } \sin\theta = \frac{1}{3}$$

First solve $\cos\theta = 0$ for $0^\circ \leq \theta < 360^\circ$:

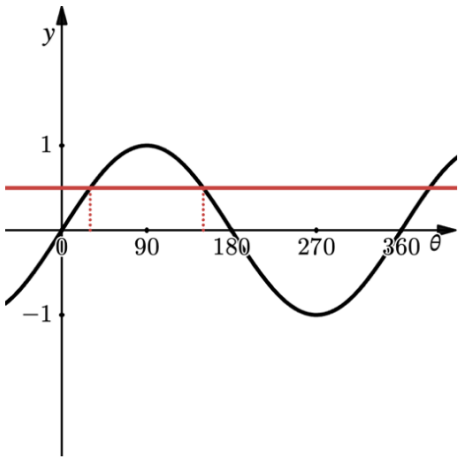


$$\theta = 90^\circ$$
$$\theta = 270^\circ$$

Now solve $\sin\theta = \frac{1}{3}$ for $0^\circ \leq \theta < 360^\circ$.

Find first value between 0° and 360°:

$$\theta = \arcsin\frac{1}{3} = 19.471\ldots^\circ \quad \text{(in range)}$$



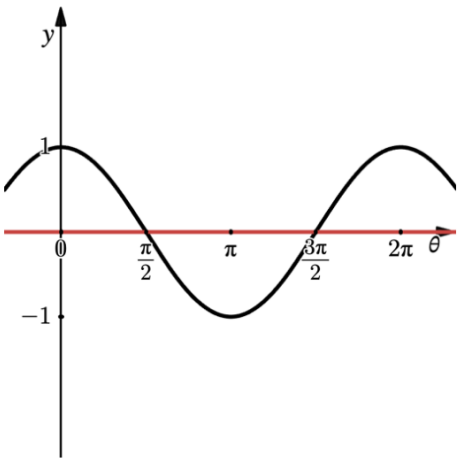
Find second value between 0° and 360°:

$$\theta = 180^\circ - 19.471\ldots^\circ = 160.528\ldots^\circ \quad \text{(in range)}$$
$$\theta = \{19.5^\circ, 90^\circ, 161^\circ, 270^\circ\} \text{ to 3 s.f.}$$

8 d)

$$8\sin 2\theta + 5\cos\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$8 \times 2\sin\theta\cos\theta + 5\cos\theta = 0$$
$$16\sin\theta\cos\theta + 5\cos\theta = 0$$
$$\cos\theta(16\sin\theta + 5) = 0$$
$$\cos\theta = 0 \text{ or } \sin\theta = -\frac{5}{16}$$

First solve $\cos\theta = 0$ for $0 \leq \theta < 2\pi$:

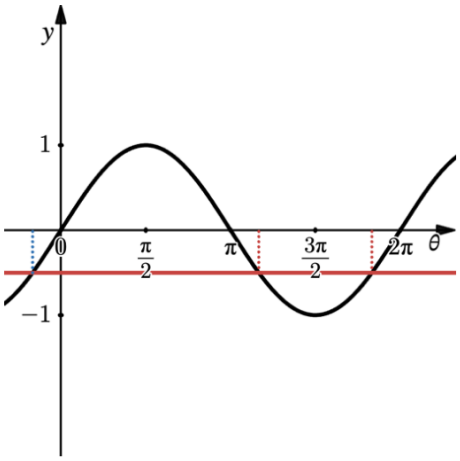


$$\theta = \frac{\pi}{2}$$
$$\theta = \frac{3\pi}{2}$$

Now solve $\sin\theta = -\frac{5}{16}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta = \arcsin(-\frac{5}{16}) = -0.3178\ldots \quad \text{(not in range)}$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.3178\ldots = 3.4594\ldots \quad \text{(in range)}$$

Find second value between $0°$ and $360°$:

$$\theta = 2\pi - 0.3178\ldots = 5.9653\ldots \quad \text{(in range)}$$
$$\theta = \{\frac{\pi}{2}, 3.46, \frac{3\pi}{2}, 5.97\} \text{ to 3 s.f.}$$

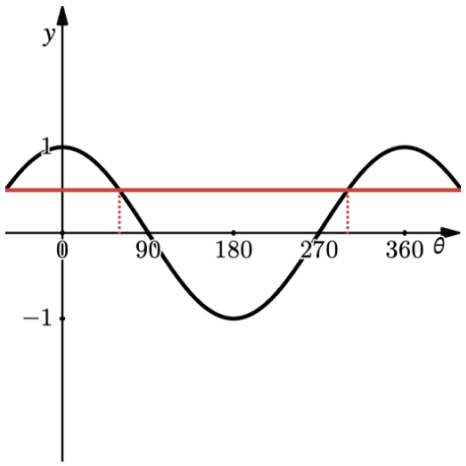
9 a)

$$\cos 2\theta + \cos\theta = 0 \text{ for } 0° \leq \theta < 360°$$
$$2\cos^2\theta - 1 + \cos\theta = 0$$
$$(2\cos\theta - 1)(\cos\theta + 1) = 0$$
$$\cos\theta = \frac{1}{2} \text{ or } \cos\theta = -1$$

First solve $\cos\theta = \frac{1}{2}$ for $0° \leq \theta < 360°$.

Find first value between $0°$ and $360°$:

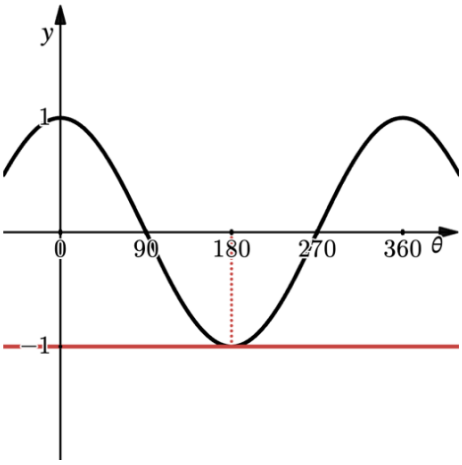
$$\theta = \arccos\frac{1}{2} = 60° \quad \text{(in range)}$$



Find second value between $0°$ and $360°$:

$$\theta = 360° - 60° = 300° \quad \text{(in range)}$$

Now solve $\cos\theta = -1$ for $0° \leq \theta < 360°$:



$$\theta = 180^\circ$$
$$\Rightarrow \theta = \{60^\circ, 180^\circ, 300^\circ\}$$

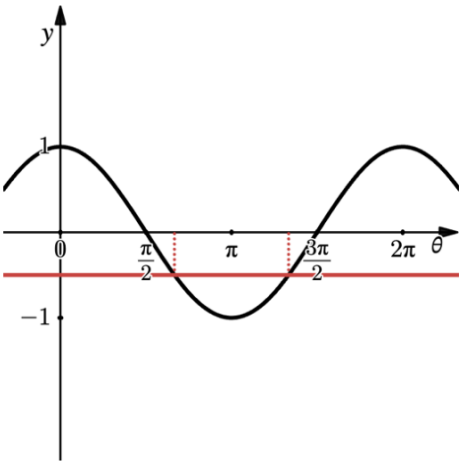
9 b)

$$3\cos 2\theta = 7\cos\theta \text{ for } 0 \leq \theta < 2\pi$$
$$3(2\cos^2\theta - 1) - 7\cos\theta = 0$$
$$6\cos^2\theta - 7\cos\theta - 3 = 0$$
$$(3\cos\theta + 1)(2\cos\theta - 3) = 0$$
$$\cos\theta = -\frac{1}{3} \text{ or } \cos\theta = \frac{3}{2}$$

First solve $\cos\theta = -\frac{1}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

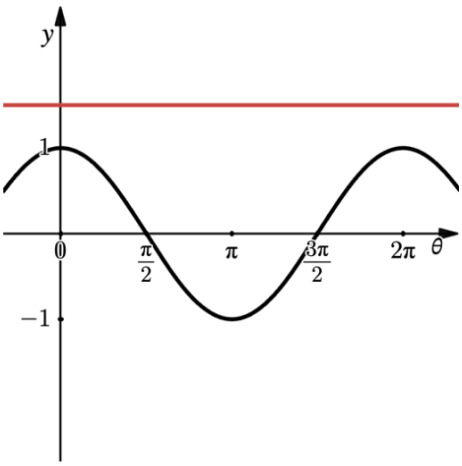
$$\theta = \arccos(-\frac{1}{3}) = 1.9106\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.9106\ldots = 4.3725\ldots \quad \text{(in range)}$$

As for $\cos\theta = \frac{3}{2}$, there are no solutions:



$$\Rightarrow \theta = \{1.91, 4.37\} \text{ to 3 s.f.}$$

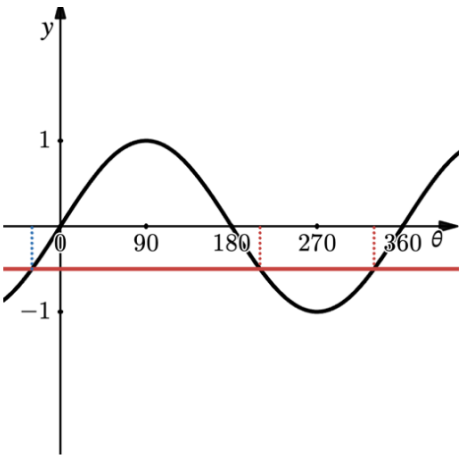
9 c)

$$\begin{aligned} 2\cos 2\theta + 2\sin\theta &= 0 \text{ for } 0^\circ \leq \theta < 360^\circ \\ 2(1 - 2\sin^2\theta) + 2\sin\theta &= 0 \\ 1 - 2\sin^2\theta + \sin\theta &= 0 \\ 2\sin^2\theta - \sin\theta - 1 &= 0 \\ (2\sin\theta + 1)(\sin\theta - 1) &= 0 \\ \sin\theta = -\frac{1}{2} \text{ or } \sin\theta &= 1 \end{aligned}$$

First solve $\sin\theta = -\frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

Find first value:

$$\theta = \arcsin(-\frac{1}{2}) = -30^\circ \quad \text{(not in range)}$$



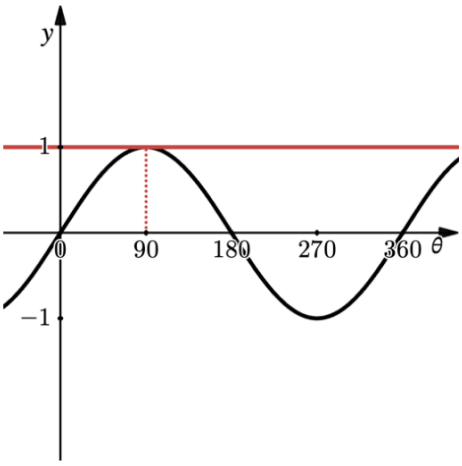
Find first value between 0° and 360°:

$$\theta = 180^\circ + 30^\circ = 210^\circ \quad \text{(in range)}$$

Find second value between 0° and 360°:

$$\theta = 360^\circ - 30^\circ = 330^\circ \quad \text{(in range)}$$

Now solve $\sin\theta = 1$ for $0^\circ \leq \theta < 360^\circ$:



$$\theta = 90°$$
$$\theta = \{90°, 210°, 330°\}$$

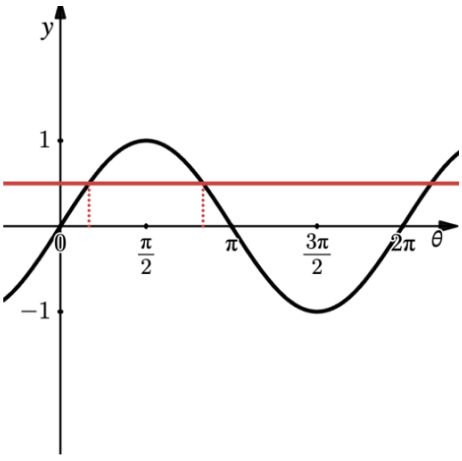
9 d)

$$2\cos 2\theta = 7\sin\theta \text{ for } 0 \leq \theta < 2\pi$$
$$2(1 - 2\sin^2\theta) = 9\sin\theta$$
$$2 - 4\sin^2\theta = 9\sin\theta$$
$$4\sin^2\theta + 9\sin\theta - 2 = 0$$
$$\sin\theta = \frac{-9 \pm \sqrt{113}}{8}$$

First solve $\sin\theta = \frac{-9+\sqrt{113}}{8}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

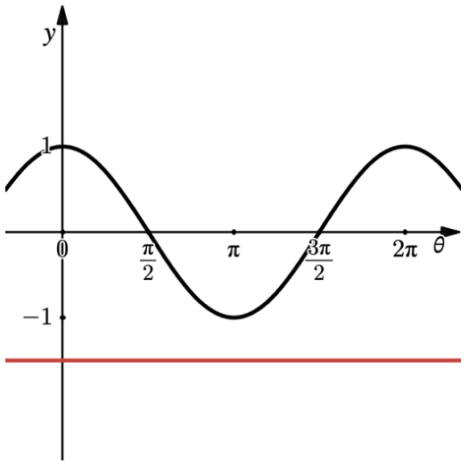
$$\theta = \arcsin(\frac{-9+\sqrt{113}}{8}) = 0.2052\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = \pi - 0.2052\ldots = 2.9363\ldots \quad \text{(in range)}$$

As for $\cos\theta = \frac{-9-\sqrt{113}}{8}$, there are no solutions:



$$\Rightarrow \theta = \{0.205, 2.94\} \text{ to 3 s.f.}$$

10 a)

$$2 \sin 2\theta \operatorname{cosec} \theta = 1 \text{ for } 0^\circ < \theta < 360^\circ$$

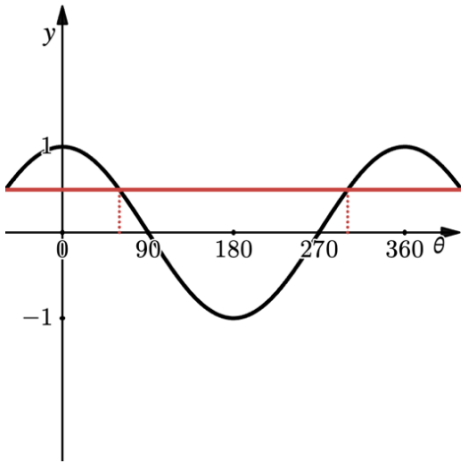
$$2 \times 2 \sin\theta \cos\theta \times \frac{1}{\sin\theta} = 1$$

$$4 \cos\theta = 1$$

$$\cos\theta = \frac{1}{4}$$

Find first value between 0° and 360°:

$$\theta = \arccos\frac{1}{4} = 75.522\ldots^\circ \quad \text{(in range)}$$



Find second value between 0° and 360°:

$$\theta = 360^\circ - 75.522\ldots^\circ = 284.477\ldots^\circ \quad \text{(in range)}$$

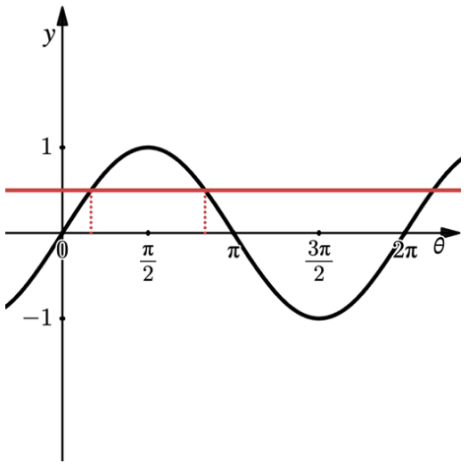
$$\Rightarrow \theta = \{75.5^\circ, 284^\circ\} \text{ to 3 s.f.}$$

10 b)

$$5\sin 2\theta \sec\theta = 2 \text{ for } 0 < \theta < 2\pi$$
$$5 \times 2\sin\theta\cos\theta \times \frac{1}{\cos\theta} = 2$$
$$10\sin\theta = 2$$
$$\sin\theta = \frac{1}{5}$$

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{1}{5} = 0.2013\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = \pi - 0.2013\ldots = 2.9402\ldots \quad \text{(in range)}$$
$$\theta = \{0.201, 2.94\} \text{ to 3 s.f.}$$

10 c)

$$\cos 2\theta \operatorname{cosec} \theta = 3 \text{ for } 0^\circ < \theta < 360^\circ$$

$$(1 - 2\sin^2\theta) \times \frac{1}{\sin\theta} = 3$$

$$1 - 2\sin^2\theta = 3\sin\theta$$

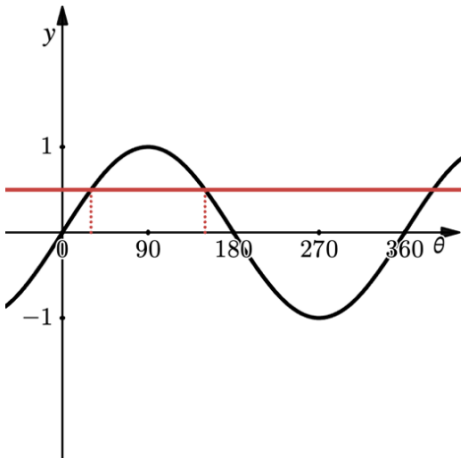
$$2\sin^2\theta + 3\sin\theta - 1 = 0$$

$$\sin\theta = \frac{-3 \pm \sqrt{17}}{4}$$

First solve $\sin\theta = \frac{-3+\sqrt{17}}{4}$ for $0^\circ < \theta < 360^\circ$.

Find first value between 0° and 360°:

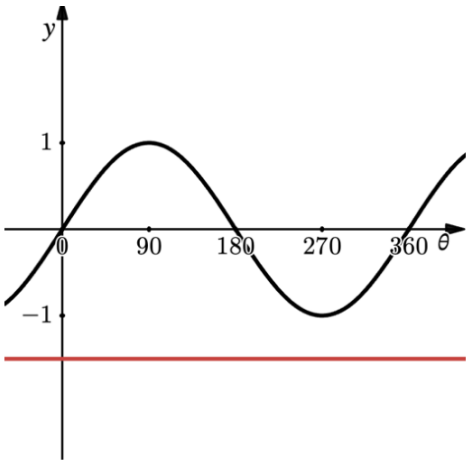
$$\theta = \arcsin\left(\frac{-3+\sqrt{17}}{4}\right) = 16.306\ldots^\circ \quad \text{(in range)}$$



Find second value between 0° and 360°:

$$\theta = 180^\circ - 16.306\ldots^\circ = 163.693\ldots^\circ \quad \text{(in range)}$$

As for $\sin\theta = \frac{-3-\sqrt{17}}{4}$, there are no solutions:



$$\theta = \{16.3^\circ, 164^\circ\} \text{ to 3 s.f.}$$

10 d)

$$9\cos 2\theta \sec\theta = 4 \text{ for } 0 < \theta < 2\pi$$

$$9(2\cos^2\theta - 1) \times \frac{1}{\cos\theta} = 4$$

$$18\cos^2\theta - 9 = 4\cos\theta$$

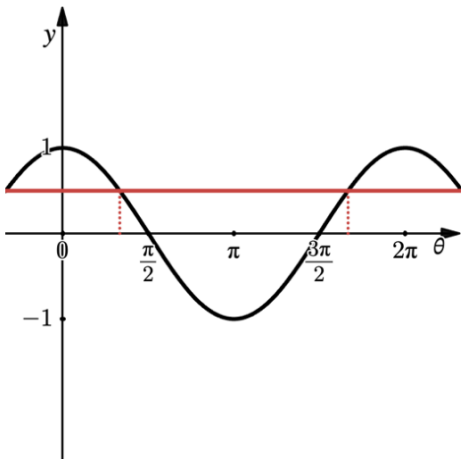
$$18\cos^2\theta - 4\cos\theta - 9 = 0$$

$$\cos\theta = \frac{2 \pm \sqrt{166}}{18}$$

First solve $\cos\theta = \frac{2+\sqrt{166}}{18}$ for $0 < \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos(\frac{2+\sqrt{166}}{18}) = 0.5972\ldots \quad \text{(in range)}$$



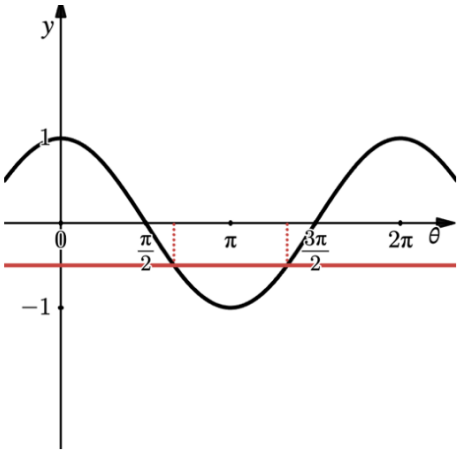
Find second value between 0 and 2π:

$$\theta = 2\pi - 0.5972\ldots = 5.6859\ldots \quad \text{(in range)}$$

Now solve $\cos\theta = \frac{2-\sqrt{166}}{18}$ for $0 < \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos(\frac{2-\sqrt{166}}{18}) = 2.2201\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

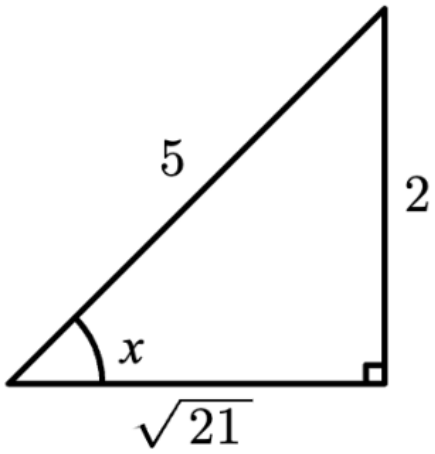
$$\theta = 2\pi - 2.2201\ldots = 4.0630\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{0.597, 2.22, 4.06, 5.69\} \text{ to 3 s.f.}$$

11)

$\sin x = \frac{2}{5}$.

x is **ACUTE**.

Draw a triangle with this information:



$$\cos x = \frac{\sqrt{21}}{5}$$
$$\Rightarrow \sin 2x = 2 \times \frac{2}{5} \times \frac{\sqrt{21}}{5} = \frac{4\sqrt{21}}{25}$$

12)

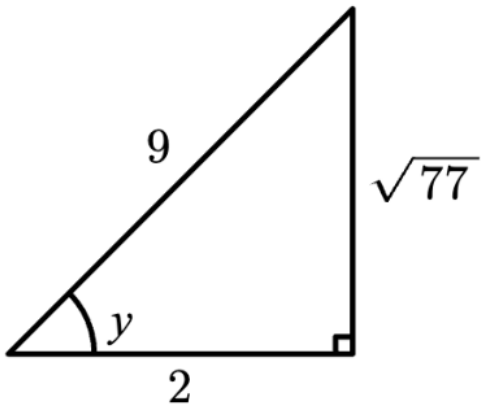
$$\sin x = \frac{3}{8}$$
$$\Rightarrow \cos 2x = 1 - 2 \times (\frac{3}{8})^2 = \frac{23}{32}$$

13)

$\cos x = -\frac{2}{9}$.

x is **OBTUSE**.

Ignore $90° < x < 180°$ and the negative sign for the moment and draw a right-angled triangle with this information:



$\tan x = -\frac{\sqrt{77}}{2}$ (the result is negative as $\tan x$ is below the x-axis between 90° and 180°).

$$\Rightarrow \tan 2x = \frac{2 \times \frac{-\sqrt{77}}{2}}{1 - (\frac{-\sqrt{77}}{2})^2} = -\frac{\sqrt{77}}{1 - \frac{77}{4}} = \frac{4\sqrt{77}}{73}$$

14)

$$\begin{aligned}
10\sin^2 x + 5\sin 2x &= 4 \\
10\sin^2 x + 5 \times 2\sin x \cos x - 4 &= 0 \\
5\sin^2 x + 5\sin x \cos x - 2 &= 0 \\
\frac{5\sin^2 x}{\cos^2 x} + \frac{5\sin x \cos x}{\cos^2 x} - \frac{2}{\cos^2 x} &= 0 \\
5\tan^2 x + 5\tan x - 2\sec^2 x &= 0 \\
5\tan^2 x + 5\tan x - 2(\tan^2 x + 1) &= 0 \\
5\tan^2 x + 5\tan x - 2\tan^2 x - 2 &= 0 \\
3\tan^2 x + 5\tan x - 2 &= 0
\end{aligned}$$

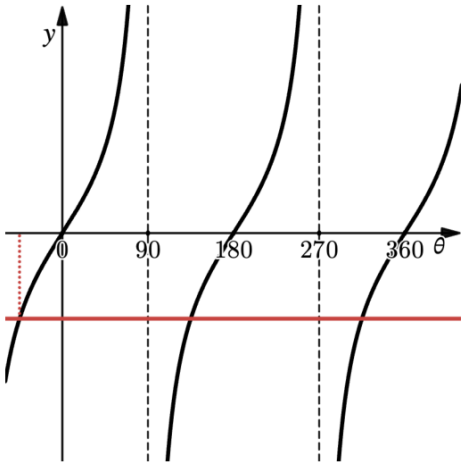
15 a)

$$
\begin{aligned}
34\sin^2\theta - \sin 2\theta &= 30 \text{ for } 0^\circ \le \theta < 360^\circ \\
34\sin^2\theta - 2\sin\theta\cos\theta - 30 &= 0 \\
17\sin^2\theta - \sin\theta\cos\theta - 15 &= 0 \\
\frac{17\sin^2\theta}{\cos^2\theta} - \frac{\sin\theta\cos\theta}{\cos^2\theta} - \frac{15}{\cos^2\theta} &= 0 \\
17\tan^2\theta - \tan\theta - 15\sec^2\theta &= 0 \\
17\tan^2\theta - \tan\theta - 15(\tan^2\theta + 1) &= 0 \\
2\tan^2\theta - \tan\theta - 15 &= 0 \\
(2\tan\theta + 5)(\tan\theta - 3) &= 0 \\
\tan\theta = -\frac{5}{2} &\text{ or } \tan\theta = 3
\end{aligned}
$$

First solve $\tan\theta = -\frac{5}{2}$ for $0^\circ \le \theta < 360^\circ$.

Find first value:

$$\theta = \arctan(-\frac{5}{2}) = -68.198\ldots^\circ \quad \text{(not in range)}$$



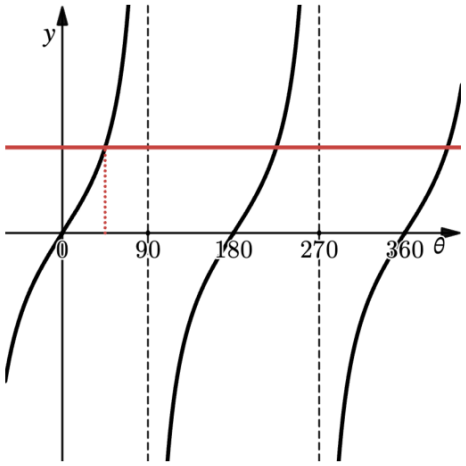
Add on period:

$$\theta = -68.198\ldots^\circ + 180^\circ = 111.801\ldots^\circ \quad \text{(in range)}$$
$$\theta = 111.801\ldots^\circ + 180^\circ = 291.801\ldots^\circ \quad \text{(in range)}$$

Now solve $\tan\theta = 3$ for $0^\circ \le \theta < 360^\circ$.

Find first value:

$$\theta = \arctan 3 = 71.565\ldots^\circ \quad \text{(in range)}$$



Add period:

$$\theta = 71.565\ldots^\circ + 180^\circ = 251.565\ldots^\circ \quad \text{(in range)}$$
$$\Rightarrow \theta = \{71.6^\circ, 112^\circ, 252^\circ, 292^\circ\} \text{ to 3 s.f.}$$

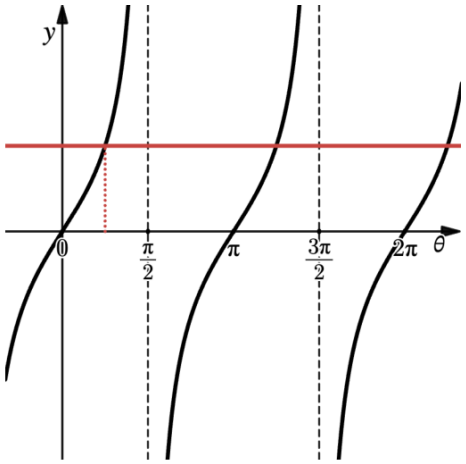
15 b)

$$22\cos^2\theta - 13\sin 2\theta = 10 \text{ for } 0 < \theta < 2\pi$$
$$22\cos^2\theta - 13 \times 2\sin\theta\cos\theta - 10 = 0$$
$$11\cos^2\theta - 13\sin\theta\cos\theta - 5 = 0$$
$$\frac{11\cos^2\theta}{\cos^2\theta} - \frac{13\sin\theta\cos\theta}{\cos^2\theta} - \frac{5}{\cos^2\theta} = 0$$
$$11 - 13\tan\theta - 5\sec^2\theta = 0$$
$$11 - 13\tan\theta - 5(\tan^2\theta + 1) = 0$$
$$6 - 13\tan\theta - 5\tan^2\theta = 0$$
$$5\tan^2\theta + 13\tan\theta - 6 = 0$$
$$(5\tan\theta - 2)(\tan\theta + 3) = 0$$
$$\tan\theta = \frac{2}{5} \text{ or } \tan\theta = -3$$

First solve $\tan\theta = \frac{2}{5}$ for $0 < \theta < 2\pi$.

Find first value:

$$\theta = \arctan\frac{2}{5} = 0.3805\ldots \quad \text{(in range)}$$



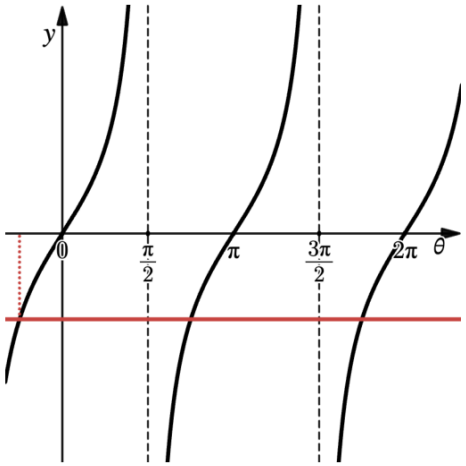
Add period:

$$\theta = 0.3805\ldots + \pi = 3.5220\ldots \quad \text{(in range)}$$

Now solve $\tan\theta = -3$ for $0 < \theta < 2\pi$.

Find first value:

$$\theta = \arctan(-3) = -1.2490\ldots \quad \text{(not in range)}$$



Add on period:

$$\theta = -1.2490\ldots + \pi = 1.8925\ldots \quad \text{(in range)}$$
$$\theta = 1.8925\ldots + \pi = 5.0341\ldots \quad \text{(in range)}$$
$$\theta = \{0.381, 1.89, 3.52, 5.03\} \text{ to 3 s.f.}$$

16 a)

LHS: $\tan 2x + \cot x \equiv \frac{2\tan x}{1-\tan^2 x} + \frac{1}{\tan x}$

$$\equiv \frac{2\tan^2 x + (1-\tan^2 x)}{(1-\tan^2 x)\tan x} \equiv \frac{\tan^2 x + 1}{(1-\tan^2 x)\tan x}$$

$$\equiv \frac{1}{\tan x} \times \frac{\sec^2 x}{1-\tan^2 x} \equiv \cot x \times \frac{\sec^2 x}{1-\tan^2 x} \times \frac{\cos^2 x}{\cos^2 x}$$

$$\equiv \cot x \times \frac{1}{\cos^2 x - \sin^2 x} \equiv \cot x \times \frac{1}{\cos 2x}$$

$$\equiv \cot x \sec 2x : \text{RHS}$$

16 b)

LHS: $\tan 2x + \sin 2x \equiv \frac{2\tan x}{1-\tan^2 x} + \frac{2\sin x \cos x}{1}$

$$\equiv \frac{2\tan x + 2\sin x \cos x(1-\tan^2 x)}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x + 2\sin x \cos x - 2\sin x \cos x \tan^2 x}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x + 2\sin x \cos x - \frac{2\sin^3 x}{\cos x}}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x + \frac{2\sin x \cos^2 x}{\cos x} - \frac{2\sin^3 x}{\cos x}}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x + \frac{2\sin x}{\cos x}\cos^2 x - \frac{2\sin x}{\cos x}\sin^2 x}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x + 2\tan x \cos^2 x - 2\tan x \sin^2 x}{1-\tan^2 x}$$

$$\equiv \frac{2\tan x}{1-\tan^2 x} \times (1 + \cos^2 x - \sin^2 x)$$

$$\equiv \tan 2x(1 + \cos^2 x - (1-\cos^2 x))$$

$$\equiv \tan 2x(2\cos^2 x) \equiv \tan 2x(2\cos^2 x - 1 + 1)$$

$$\equiv \tan 2x(\cos 2x + 1) \equiv \tan 2x(\cos 2x + 1) : \text{RHS}$$

16 c)

LHS: $\tan 2x + \cot 2x \equiv \tan 2x + \frac{1}{\tan 2x}$

$$\equiv \frac{2\tan x}{1-\tan^2 x} + \frac{1-\tan^2 x}{2\tan x} \equiv \frac{4\tan^2 x + (1-\tan^2 x)^2}{2\tan x(1-\tan^2 x)}$$

$$\equiv \frac{4\tan^2 x + 1 - 2\tan^2 x + \tan^4 x}{2\tan x(1-\tan^2 x)} \equiv \frac{\tan^4 x + 2\tan^2 x + 1}{2\tan x(1-\tan^2 x)}$$

$$\equiv \frac{(\tan^2 x + 1)^2}{2\tan x(1-\tan^2 x)} \equiv \frac{(\sec^2 x)^2}{2\tan x(1-\tan^2 x)}$$

$$\equiv \frac{\sec^4 x}{2\tan x - 2\tan^3 x} \equiv \frac{1}{\cos^4 x} \times \frac{1}{\frac{2\sin x}{\cos x} - \frac{2\sin^3 x}{\cos^3 x}}$$

$$\equiv \frac{1}{2\sin x\cos^3 x - 2\sin^3 x\cos x} \equiv \frac{1}{2\sin x\cos x(\cos^2 x - \sin^2 x)}$$

$$\equiv \frac{1}{\sin 2x\cos 2x} \equiv \frac{2}{2\sin 2x\cos 2x} \equiv \frac{2}{\sin 4x}$$

$$\equiv 2\operatorname{cosec} 4x : \text{RHS}$$

17 a)

LHS: $\cot 2x - \cos 2x \equiv \frac{1}{\tan 2x} - \cos 2x$

$$\equiv \frac{1-\tan^2 x}{2\tan x} - \frac{\cos^2 x - \sin^2 x}{1} \equiv \frac{1-\tan^2 x - 2\tan x(\cos^2 x - \sin^2 x)}{2\tan x}$$

$$\equiv \frac{1-\tan^2 x - \frac{2\sin x}{\cos x}(\cos^2 x - \sin^2 x)}{2\tan x}$$

$$\equiv \frac{1-\tan^2 x - 2\sin x\cos x + \frac{2\sin^3 x}{\cos x}}{2\tan x}$$

$$\equiv \frac{1}{2\tan x} \times (1-\tan^2 x - 2\sin x\cos x + \frac{2\sin^3 x}{\cos x})$$

$$\equiv \frac{1-\tan^2 x}{2\tan x} \times \left(1 - \frac{2\sin x\cos x - \frac{2\sin^3 x}{\cos x}}{1-\tan^2 x}\right)$$

$$\equiv \cot 2x \left(1 - \frac{2\sin x\cos x - \frac{2\sin^3 x}{\cos x}}{1 - \frac{\sin^2 x}{\cos^2 x}}\right)$$

$$\equiv \cot 2x \left(1 - \frac{2\sin x\cos^3 x - 2\sin^3 x\cos x}{\cos^2 x - \sin^2 x}\right)$$

$$\equiv \cot 2x \left(1 - \frac{2\sin x\cos x(\cos^2 x - \sin^2 x)}{\cos^2 x - \sin^2 x}\right)$$

$$\equiv \cot 2x(1 - 2\sin x\cos x)$$

$$\equiv \cot 2x(1 - \sin 2x) : \text{RHS}$$

17 b)

LHS: $\cot 2x + \operatorname{cosec} 2x \equiv \frac{1}{\tan 2x} + \frac{1}{\sin 2x}$

$$\equiv \frac{1-\tan^2 x}{2\tan x} + \frac{1}{2\sin x\cos x} \equiv \frac{\sin x\cos x(1-\tan^2 x)}{2\sin x\cos x\tan x} + \frac{1}{2\sin x\cos x}$$

$$\equiv \frac{1}{2\sin x\cos x} \times \left(\frac{\sin x\cos x(1-\tan^2 x)}{\tan x} + 1\right)$$

$$\equiv \frac{1}{\sin 2x} \times \left(\frac{\sin x\cos x(1-\frac{\sin^2 x}{\cos^2 x})}{\frac{\sin x}{\cos x}} + 1\right)$$

$$\equiv \operatorname{cosec} 2x\left(1 + \frac{\sin x\cos x - \frac{\sin^3 x}{\cos x}}{\frac{\sin x}{\cos x}}\right)$$

$$\equiv \operatorname{cosec} 2x\left(1 + \frac{\sin x\cos^2 x - \sin^3 x}{\sin x}\right)$$

$$\equiv \operatorname{cosec} 2x\left(1 + \frac{\sin x\cos^2 x}{\sin x} - \frac{\sin^3 x}{\sin x}\right)$$

$$\equiv \operatorname{cosec} 2x(1 + \cos^2 x - \sin^2 x)$$

$$\equiv \operatorname{cosec} 2x(1 + \cos 2x) : \text{RHS}$$

17 c)

LHS: $\cot 2x + \operatorname{cosec} 2x \equiv \frac{1}{\tan 2x} + \frac{1}{\sin 2x}$

$$\equiv \frac{1-\tan^2 x}{2\tan x} + \frac{1}{2\sin x\cos x} \equiv \frac{2\sin x\cos x(1-\tan^2 x) + 2\tan x}{4\sin x\cos x\tan x}$$

$$\equiv \frac{2\sin x\cos x(1-\frac{\sin^2 x}{\cos^2 x}) + \frac{2\sin x}{\cos x}}{4\sin x\cos x\frac{\sin x}{\cos x}} \equiv \frac{2\sin x\cos x - \frac{2\sin^3 x}{\cos x} + \frac{2\sin x}{\cos x}}{4\sin^2 x}$$

$$\equiv \frac{2\sin x\cos^2 x - 2\sin^3 x + 2\sin x}{4\sin^2 x\cos x} \equiv \frac{\cos^2 x - \sin^2 x + 1}{2\sin x\cos x}$$

$$\equiv \frac{\cos^2 x - \sin^2 x + \sin^2 x + \cos^2 x}{2\sin x\cos x} \equiv \frac{2\cos^2 x}{2\sin x\cos x}$$

$$\equiv \frac{\cos x}{\sin x} \equiv \cot x : \text{RHS}$$

18)

LHS: $\tan 2x + \operatorname{cosec} 2x \equiv \frac{2\tan x}{1-\tan^2 x} + \frac{1}{\sin 2x}$

$$\equiv \frac{2\tan x}{1-\tan^2 x} + \frac{1}{2\sin x\cos x} \equiv \frac{4\sin x\cos x\tan x + 1 - \tan^2 x}{2\sin x\cos x(1-\tan^2 x)}$$

$$\equiv \frac{4\sin x\cos x\frac{\sin x}{\cos x} + 1 - \frac{\sin^2 x}{\cos^2 x}}{2\sin x\cos x(1-\frac{\sin^2 x}{\cos^2 x})} \equiv \frac{4\sin^2 x + 1 - \frac{\sin^2 x}{\cos^2 x}}{2\sin x\cos x - \frac{2\sin^3 x}{\cos x}}$$

$$\equiv \frac{4\sin^2 x\cos^2 x + \cos^2 x - \sin^2 x}{2\sin x\cos^3 x - 2\sin^3 x\cos x} \equiv \frac{(2\sin x\cos x)^2 + \cos 2x}{2\sin x\cos x(\cos^2 x - \sin^2 x)}$$

$$\equiv \frac{(\sin 2x)^2 + \cos 2x}{\sin 2x\cos 2x} \equiv \frac{2\sin^2 2x + 2\cos 2x}{2\sin 2x\cos 2x} \equiv \frac{2\sin^2 2x + 2\cos 2x}{\sin 4x}$$

$$\equiv \frac{2}{\sin 4x} \times (\sin^2 2x + \cos 2x)$$

$$\equiv 2\operatorname{cosec} 4x(\sin^2 2x + \cos 2x) : \text{RHS}$$

19 a)

LHS: $\frac{2\tan 2x}{1+\tan^2 2x} \equiv \frac{\frac{2\sin 2x}{\cos 2x}}{1+\frac{\sin^2 2x}{\cos^2 2x}} \equiv \frac{2\sin 2x\cos 2x}{\cos^2 2x+\sin^2 2x} \equiv \frac{\sin 4x}{1}$

$$\equiv \sin 4x : \text{RHS}$$

19 b)

LHS: $(\sin 2x + \cos 2x)^2$

$$\equiv \sin^2 2x + 2\sin 2x\cos 2x + \cos^2 2x$$

$$\equiv (\sin^2 2x + \cos^2 2x) + 2\sin 2x\cos 2x$$

$$\equiv 1 + \sin 4x : \text{RHS}$$

19 c)

LHS: $\frac{2\sin x}{\cos 2x+1} - \frac{2\sin x}{\cos 2x-1} \equiv \frac{2\sin x(\cos 2x-1)-2\sin x(\cos 2x+1)}{(\cos 2x+1)(\cos 2x-1)}$

$$\equiv \frac{2\sin x(\cos 2x - 1 - \cos 2x - 1)}{\cos^2 2x - 1} \equiv \frac{-4\sin x}{\cos^2 2x - 1} \equiv \frac{4\sin x}{1 - \cos^2 2x}$$

$$\equiv \frac{4\sin x}{\sin^2 2x} \equiv \frac{4\sin x}{(2\sin x\cos x)^2} \equiv \frac{4\sin x}{4\sin^2 x\cos^2 x} \equiv \frac{1}{\sin x\cos^2 x}$$

$$\equiv \frac{1}{\sin x} \times \frac{1}{\cos^2 x} \equiv \operatorname{cosec} x\sec^2 x : \text{RHS}$$

20 a)

RHS: $\frac{2}{(\sin x+\cos x)^2}-1 \equiv \frac{2}{\sin^2 x+2\sin x\cos x+\cos^2 x}-1$

$$\equiv \frac{2}{1+2\sin x\cos x}-1 \equiv \frac{2}{1+\sin 2x}-1 \equiv \frac{2-(1+\sin 2x)}{1+\sin 2x}$$

$$\equiv \frac{2-1-\sin 2x}{1+\sin 2x} \equiv \frac{1-\sin 2x}{1+\sin 2x} : \text{LHS}$$

20 b)

RHS: $\frac{2\sin^2 x}{(\sin x+\cos x)^2} \equiv \frac{2\sin^2 x-1+1}{\sin^2 x+2\sin x\cos x+\cos^2 x}$

$$\equiv \frac{1-(1-2\sin^2 x)}{1+2\sin x\cos x} \equiv \frac{1-\cos 2x}{1+\sin 2x} : \text{LHS}$$

20 c)

RHS: $\frac{\cos x(\cot x-1)}{\sin x+\cos x} \equiv \frac{\cos x(\frac{\cos x}{\sin x}-1)}{\sin x+\cos x} \equiv \frac{\frac{\cos^2 x}{\sin x}-\cos x}{\sin x+\cos x}$

$$\equiv \frac{\cos^2 x-\sin x\cos x}{\sin^2 x+\sin x\cos x} \equiv \frac{2\cos^2 x-2\sin x\cos x}{2\sin^2 x+2\sin x\cos x} \equiv \frac{2\cos^2 x-\sin 2x}{2\sin^2 x+\sin 2x}$$

$$\equiv \frac{2\cos^2 x-1+1-\sin 2x}{2\sin^2 x-1+1+\sin 2x} \equiv \frac{1+(2\cos^2 x-1)-\sin 2x}{1+\sin 2x-(1-2\sin^2 x)}$$

$$\equiv \frac{1+\cos 2x-\sin 2x}{1+\sin 2x-\cos 2x} : \text{LHS}$$

21 a)

RHS: $\frac{\cos 2x+3\sin^2 x+\cos^2 x}{2\sin^2 x} \equiv \frac{2\cos^2 x-1+3\sin^2 x+\cos^2 x}{2\sin^2 x}$

$$\equiv \frac{3\cos^2 x+3\sin^2 x-1}{2\sin^2 x} \equiv \frac{3(\cos^2 x+\sin^2 x)-1}{2\sin^2 x} \equiv \frac{3-1}{2\sin^2 x}$$

$$\equiv \frac{2}{2\sin^2 x} \equiv \frac{1}{\sin^2 x} \equiv \operatorname{cosec}^2 x : \text{LHS}$$

21 b)

$(\cot^2\theta - 1)\operatorname{cosec}^2\theta = \frac{\cos 2\theta + 3\sin^2\theta + \cos^2\theta}{6\sin^2\theta}$ for $0 < \theta < 2\pi$.

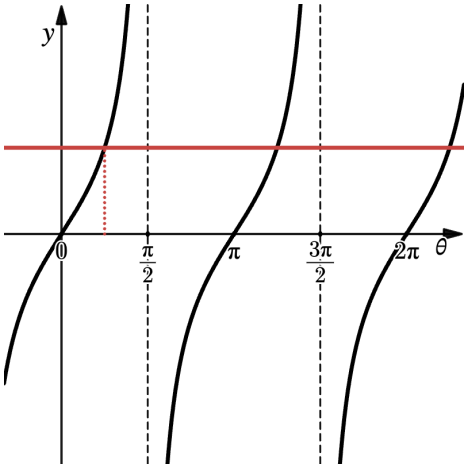
$$\Rightarrow 3(\cot^2\theta - 1) \times \frac{\cos 2\theta + 3\sin^2\theta + \cos^2\theta}{2\sin^2\theta} = \frac{\cos 2\theta + 3\sin^2\theta + \cos^2\theta}{2\sin^2\theta}$$

$$\Rightarrow 3(\cot^2\theta - 1) = 1 \Rightarrow \cot^2\theta - 1 = \frac{1}{3}$$

$$\Rightarrow \cot^2\theta = \frac{4}{3} \Rightarrow \cot\theta = \pm\frac{2}{\sqrt{3}} \Rightarrow \tan\theta = \pm\frac{\sqrt{3}}{2}$$

First solve $\tan\theta = \frac{\sqrt{3}}{2}$ for $0 < \theta < 2\pi$.

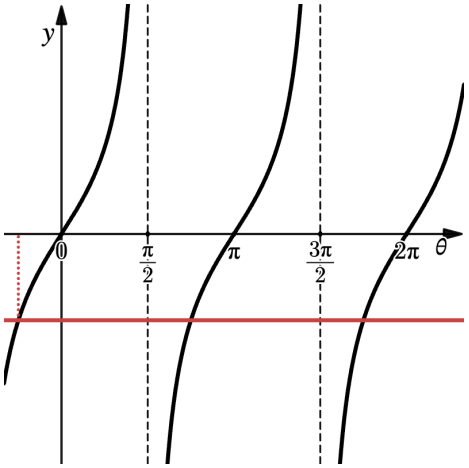
Find first value: $\theta = \arctan\frac{\sqrt{3}}{2} = 0.7137\ldots$ (in range).



Add period: $\theta = 0.7137\ldots + \pi = 3.8553\ldots$ (in range).

Now solve $\tan\theta = -\frac{\sqrt{3}}{2}$ for $0 < \theta < 2\pi$.

Find first value: $\theta = \arctan(-\frac{\sqrt{3}}{2}) = -0.7137\ldots$ (not in range).



Add on period:

$$\theta = -0.7137\ldots + \pi = 2.4278\ldots \quad \text{(in range)}$$

$$\theta = 2.4278\ldots + \pi = 5.5694\ldots \quad \text{(in range)}$$

$\theta = \{0.714, 2.43, 3.86, 5.57\}$ to 3sf.

22 a)

$11 \sin 2x = 2 \cos x + 6 \cos^3 x$ for $0 < x < 2\pi$.

$$\Rightarrow 11 \times 2 \sin x \cos x = 2 \cos x + 6 \cos^3 x$$

$$\Rightarrow 11 \sin x \cos x - \cos x - 3 \cos^3 x = 0$$

$$\Rightarrow \cos x(11 \sin x - 1 - 3 \cos^2 x) = 0$$

$$\Rightarrow \cos x(11 \sin x - 1 - 3(1 - \sin^2 x)) = 0$$

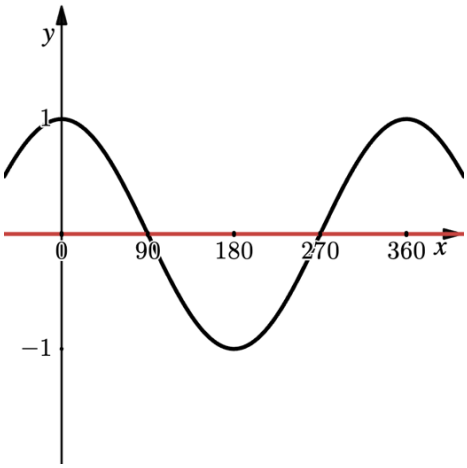
$$\Rightarrow \cos x(11 \sin x - 1 - 3 + 3 \sin^2 x) = 0$$

$$\Rightarrow \cos x(3 \sin^2 x + 11 \sin x - 4) = 0$$

$$\Rightarrow \cos x(3 \sin x - 1)(\sin x + 4) = 0$$

$$\Rightarrow \cos x = 0 \text{ or } \sin x = \frac{1}{3} \text{ or } \sin x = -4$$

First solve $\cos x = 0$ for $0 < x < 2\pi$.

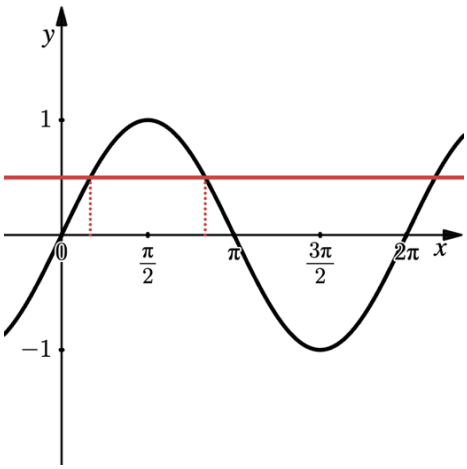


$x = \frac{\pi}{2}$

$x = \frac{3\pi}{2}$

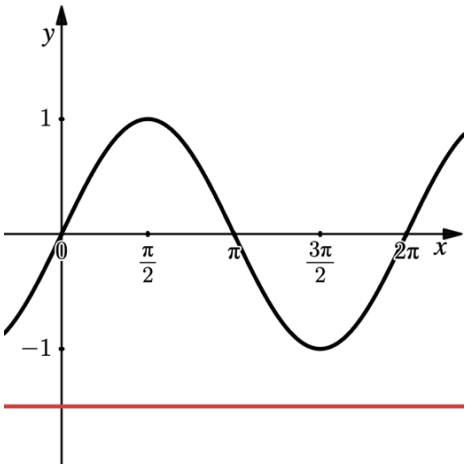
Now solve $\sin x = \frac{1}{3}$ for $0 < x < 2\pi$.

Find first value between 0 and 2π: $x = \arcsin \frac{1}{3} = 0.3398\ldots$ (in range).



Find second value between 0 and 2π: $\theta = \pi - 0.3398\ldots = 2.8017\ldots$ (in range).

As for $\sin\theta = -4$, there are no solutions:



$\Rightarrow x = \{0.340, \frac{\pi}{2}, 2.80, \frac{3\pi}{2}\}$ to 3sf.

22 b)

$$3\theta - \frac{\pi}{4} = 0.3398\ldots \Rightarrow \theta = 0.3750\ldots$$

$$3\theta - \frac{\pi}{4} = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$$

$$3\theta - \frac{\pi}{4} = 2.8017\ldots \Rightarrow \theta = 1.1957\ldots$$

$$3\theta - \frac{\pi}{4} = \frac{3\pi}{2} \Rightarrow \theta = \frac{7\pi}{12}$$

$\Rightarrow \theta = \{0.375, \frac{\pi}{4}, 1.20, \frac{7\pi}{12}\}$ to 3sf.

23 a)

LHS: $\sqrt{2}\sin(2x + \frac{\pi}{4})$

$$\equiv \sqrt{2}(\sin 2x \cos\tfrac{\pi}{4} + \cos 2x \sin\tfrac{\pi}{4})$$

$$\equiv \sqrt{2}(\tfrac{\sqrt{2}}{2}\sin 2x + \tfrac{\sqrt{2}}{2}\cos 2x) \equiv \sin 2x + \cos 2x$$

$$\equiv 2\sin x \cos x + (\cos^2 x - \sin^2 x)$$

$$\equiv \cos^2 x + 2\sin x \cos x - \sin^2 x : \text{RHS}$$

23 b)

LHS: $2\sqrt{3}\sin(\frac{\pi}{3} - 2x)$

$$\equiv 2\sqrt{3}(\sin\tfrac{\pi}{3}\cos 2x - \cos\tfrac{\pi}{3}\sin 2x)$$
$$\equiv 2\sqrt{3}(\tfrac{\sqrt{3}}{2}\cos 2x - \tfrac{1}{2}\sin 2x)$$
$$\equiv 3\cos 2x - \sqrt{3}\sin 2x$$
$$\equiv 3(\cos^2 x - \sin^2 x) - \sqrt{3} \times 2\sin x\cos x$$
$$\equiv 3\cos^2 x - 2\sqrt{3}\sin x\cos x - 3\sin^2 x : \text{RHS}$$

23 c)

LHS: $\sin(\frac{\pi}{4} - 2x) + \cos(2x + \frac{\pi}{4}) + 2\sqrt{2}\sin^2 x$

$$\equiv (\sin\tfrac{\pi}{4}\cos 2x - \cos\tfrac{\pi}{4}\sin 2x) + (\cos 2x\cos\tfrac{\pi}{4} - \sin 2x\sin\tfrac{\pi}{4}) + 2\sqrt{2}\sin^2 x$$
$$\equiv (\tfrac{\sqrt{2}}{2}\cos 2x - \tfrac{\sqrt{2}}{2}\sin 2x) + (\tfrac{\sqrt{2}}{2}\cos 2x - \tfrac{\sqrt{2}}{2}\sin 2x) + 2\sqrt{2}\sin^2 x$$
$$\equiv \sqrt{2}\cos 2x - \sqrt{2}\sin 2x + 2\sqrt{2}\sin^2 x$$
$$\equiv \sqrt{2}(\cos^2 x - \sin^2 x) - \sqrt{2}(2\sin x\cos x) + 2\sqrt{2}\sin^2 x$$
$$\equiv \sqrt{2}(\cos^2 x - \sin^2 x - 2\sin x\cos x + 2\sin^2 x)$$
$$\equiv \sqrt{2}(\cos^2 x - 2\sin x\cos x + \sin^2 x)$$
$$\equiv \sqrt{2}(\cos x - \sin x)^2 : \text{RHS}$$

23 d)

LHS: $\frac{1}{2}\cos(\frac{\pi}{6} - 2x) - \frac{1}{2}\sin(\frac{\pi}{3} - 2x)$

$$\equiv \tfrac{1}{2}(\cos\tfrac{\pi}{6}\cos 2x + \sin\tfrac{\pi}{6}\sin 2x) - \tfrac{1}{2}(\sin\tfrac{\pi}{3}\cos 2x - \cos\tfrac{\pi}{3}\sin 2x)$$
$$\equiv \tfrac{1}{2}(\tfrac{\sqrt{3}}{2}\cos 2x + \tfrac{1}{2}\sin 2x) - \tfrac{1}{2}(\tfrac{\sqrt{3}}{2}\cos 2x - \tfrac{1}{2}\sin 2x)$$
$$\equiv \tfrac{\sqrt{3}}{4}\cos 2x + \tfrac{1}{4}\sin 2x - \tfrac{\sqrt{3}}{4}\cos 2x + \tfrac{1}{4}\sin 2x$$
$$\equiv \tfrac{1}{2}\sin 2x \equiv \tfrac{1}{2} \times 2\sin x\cos x \equiv \sin x\cos x : \text{RHS}$$

24)

LHS: $(\sin(x + \frac{\pi}{4}) + \cos(x + \frac{\pi}{4}))^2$

$$\equiv (\sin x\cos\tfrac{\pi}{4} + \cos x\sin\tfrac{\pi}{4} + \cos x\cos\tfrac{\pi}{4} - \sin x\sin\tfrac{\pi}{4})^2$$
$$\equiv (\tfrac{\sqrt{2}}{2}\sin x + \tfrac{\sqrt{2}}{2}\cos x + \tfrac{\sqrt{2}}{2}\cos x - \tfrac{\sqrt{2}}{2}\sin x)^2$$
$$\equiv (\sqrt{2}\cos x)^2 \equiv 2\cos^2 x \equiv 2\cos^2 x - 1 + 1$$
$$\equiv \cos 2x + 1 : \text{RHS}$$

25)

LHS: $(\sin(2x+\frac{\pi}{4})-\cos(2x+\frac{\pi}{4}))^2$

$$\equiv ((\sin 2x\cos\tfrac{\pi}{4}+\cos 2x\sin\tfrac{\pi}{4})-(\cos 2x\cos\tfrac{\pi}{4}-\sin 2x\sin\tfrac{\pi}{4}))^2$$

$$\equiv (\tfrac{\sqrt{2}}{2}\sin 2x+\tfrac{\sqrt{2}}{2}\cos 2x-\tfrac{\sqrt{2}}{2}\cos 2x+\tfrac{\sqrt{2}}{2}\sin 2x)^2$$

$$\equiv (\sqrt{2}\sin 2x)^2 \equiv 2\sin^2 2x \equiv 1+(2\sin^2 2x-1)$$

$$\equiv 1-(1-2\sin^2 2x) \equiv 1-\cos 4x : \text{RHS}$$

26)

$\sin 3x \equiv \sin(2x+x) \equiv \sin 2x\cos x+\cos 2x\sin x$

$$\equiv (2\sin x\cos x)\cos x+(1-2\sin^2 x)\sin x$$

$$\equiv 2\sin x\cos^2 x+\sin x-2\sin^3 x$$

$$\equiv 2\sin x(1-\sin^2 x)+\sin x-2\sin^3 x$$

$$\equiv 2\sin x-2\sin^3 x+\sin x-2\sin^3 x$$

$$\equiv 3\sin x-4\sin^3 x$$

27)

$\cos 3x \equiv \cos(2x+x) \equiv \cos 2x\cos x-\sin 2x\sin x$

$$\equiv (2\cos^2 x-1)\cos x-(2\sin x\cos x)\sin x$$

$$\equiv 2\cos^3 x-\cos x-2\sin^2 x\cos x$$

$$\equiv 2\cos^3 x-\cos x-2(1-\cos^2 x)\cos x$$

$$\equiv 2\cos^3 x-\cos x-2\cos x+2\cos^3 x$$

$$\equiv 4\cos^3 x-3\cos x$$

28)

LHS: $\sqrt{2}\sin(\frac{\pi}{4}-x)(2\sin 2x+1)$

$$\equiv \sqrt{2}(\sin\tfrac{\pi}{4}\cos x-\cos\tfrac{\pi}{4}\sin x)(2\times 2\sin x\cos x+1)$$

$$\equiv \sqrt{2}(\tfrac{\sqrt{2}}{2}\cos x-\tfrac{\sqrt{2}}{2}\sin x)(4\sin x\cos x+1)$$

$$\equiv (\cos x-\sin x)(4\sin x\cos x+1)$$

$$\equiv 4\sin x\cos^2 x+\cos x-4\sin^2 x\cos x-\sin x$$

$$\equiv 4\sin x(1-\sin^2 x)+\cos x-4(1-\cos^2 x)\cos x-\sin x$$

$$\equiv 4\sin x-4\sin^3 x+\cos x-4\cos x+4\cos^3 x-\sin x$$

$$\equiv (3\sin x-4\sin^3 x)+(4\cos^3 x-3\cos x)$$

$$\equiv \sin 3x+\cos 3x : \text{RHS}$$

29)

$\cos 2x \equiv 2\cos^2 x - 1 \Rightarrow \cos^2 x \equiv \frac{1}{2}(\cos 2x + 1)$.

$$\mathrm{f}(x) = 5 - \cos^2 x = 5 - \tfrac{1}{2}(\cos 2x + 1)$$
$$\Rightarrow \mathrm{f}(x) = \tfrac{9}{2} - \tfrac{1}{2}\cos 2x$$

$\therefore \mathrm{f}(x)$ has period π (or $180°$).

30)

LHS: $\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} \equiv \frac{3\sin x - 4\sin^3 x}{\sin x} + \frac{4\cos^3 x - 3\cos x}{\cos x}$

$$\equiv \frac{3\sin x}{\sin x} - \frac{4\sin^3 x}{\sin x} + \frac{4\cos^3 x}{\cos x} - \frac{3\cos x}{\cos x}$$
$$\equiv 3 - 4\sin^2 x + 4\cos^2 x - 3 \equiv 4(\cos^2 x - \sin^2 x)$$
$$\equiv 4\cos 2x : \text{RHS}$$

31)

$\tan 3x \equiv \tan(2x + x) \equiv \frac{\tan 2x + \tan x}{1 - \tan 2x \tan x}$

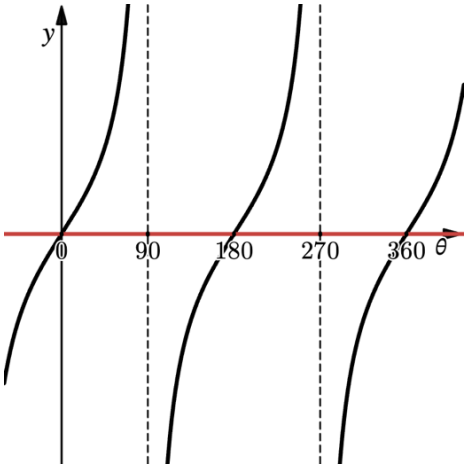
$$\equiv \frac{\frac{2\tan x}{1-\tan^2 x} + \tan x}{1 - \frac{2\tan x}{1-\tan^2 x}\tan x} \equiv \frac{2\tan x + \tan x(1 - \tan^2 x)}{(1 - \tan^2 x) - 2\tan x \times \tan x}$$
$$\equiv \frac{2\tan x + \tan x - \tan^3 x}{1 - \tan^2 x - 2\tan^2 x} \equiv \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$$

32)

$\tan 3\theta + \tan\theta = 0$ for $0° < \theta < 360°$.

$$\Rightarrow \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} + \tan\theta = 0$$
$$\Rightarrow 3\tan\theta - \tan^3\theta + \tan\theta(1 - 3\tan^2\theta) = 0$$
$$\Rightarrow 3\tan\theta - \tan^3\theta + \tan\theta - 3\tan^3\theta = 0$$
$$\Rightarrow 4\tan\theta - 4\tan^3\theta = 0 \Rightarrow \tan\theta - \tan^3\theta = 0$$
$$\Rightarrow \tan\theta(1 - \tan^2\theta) = 0$$
$$\Rightarrow \tan\theta(1 - \tan\theta)(1 + \tan\theta) = 0$$
$$\Rightarrow \tan\theta = 0 \text{ or } \tan\theta = 1 \text{ or } \tan\theta = -1$$

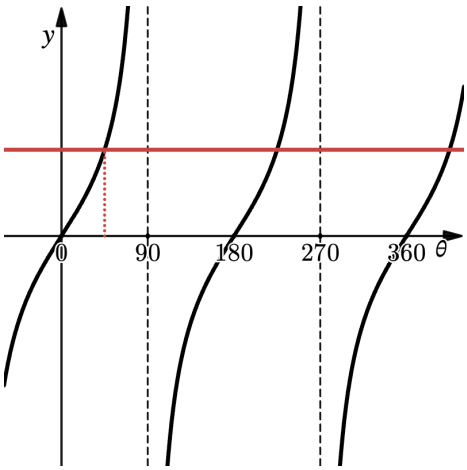
First solve $\tan\theta = 0$ for $0° < \theta < 360°$.



$\theta = 180°$.

Now solve $\tan\theta = 1$ for $0° < \theta < 360°$.

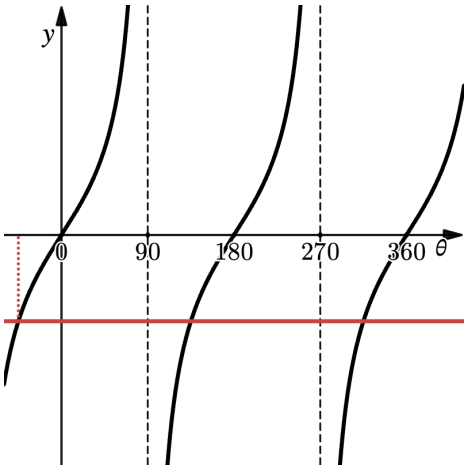
Find first value: $\theta = \arctan 1 = 45°$ (in range).



Add period: $\theta = 45° + 180° = 225°$ (in range).

Now solve $\tan\theta = -1$ for $0° < \theta < 360°$.

Find first value: $\theta = \arctan(-1) = -45°$ (not in range).



Add on period:

$$\theta = -45° + 180° = 135° \quad \text{(in range)}$$

$$\theta = 135° + 180° = 315° \quad \text{(in range)}$$

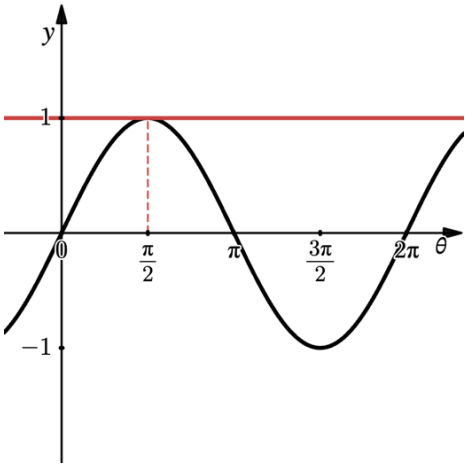
$\Rightarrow \theta = \{45°, 135°, 180°, 225°, 315°\}$.

33)

$\sin\frac{2\theta}{3}\cos\frac{2\theta}{3} = \frac{1}{2}$ for $0 < \theta < 2\pi$.

$$\Rightarrow 2\sin\tfrac{2\theta}{3}\cos\tfrac{2\theta}{3} = 1 \Rightarrow \sin(2\times\tfrac{2\theta}{3}) = 1 \Rightarrow \sin\tfrac{4\theta}{3} = 1$$

Find first value between 0 and 2π: $\frac{4\theta}{3} = \arcsin 1 = \frac{\pi}{2}$.



Apply transformation: $\frac{4\theta}{3} = \frac{\pi}{2} \Rightarrow \theta = \frac{3\pi}{8}$ (in range).

Add period: $\theta = \frac{3\pi}{8} + \frac{3\pi}{2} = \frac{15\pi}{8}$ (in range).

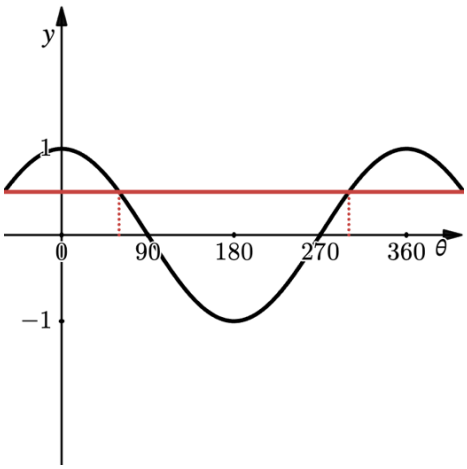
$\theta = \{\frac{3\pi}{8}, \frac{15\pi}{8}\}$.

34)

$\cos^2\frac{3\theta}{2} - \sin^2\frac{3\theta}{2} = \frac{1}{3}$ for $0° < \theta < 360°$.

$$\Rightarrow \cos(2\times\tfrac{3\theta}{2}) = \tfrac{1}{3} \Rightarrow \cos 3\theta = \tfrac{1}{3}$$

Find first value between 0° and 360°: $3\theta = \arccos\frac{1}{3} = 70.528\ldots°$.



Find second value between 0° and 360°: $3\theta = 360° - 70.528\ldots° = 289.471\ldots°$.

Apply transformation:

$$3\theta = 70.528\ldots° \Rightarrow \theta = 23.509\ldots° \quad \text{(in range)}$$

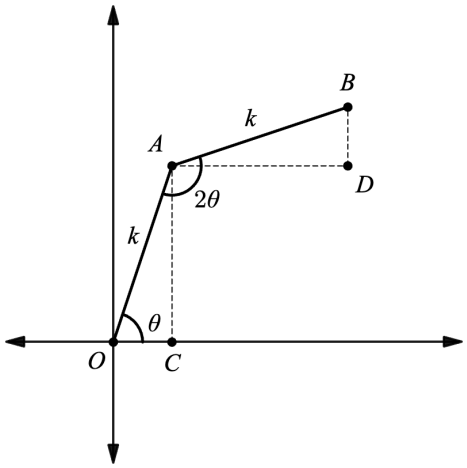
$$3\theta = 289.471\ldots° \Rightarrow \theta = 96.490\ldots° \quad \text{(in range)}$$

Add period:

$$23.509\ldots^\circ + 120^\circ = 143.509\ldots^\circ \quad \text{(in range)}$$
$$143.509\ldots^\circ + 120^\circ = 263.509\ldots^\circ \quad \text{(in range)}$$
$$96.490\ldots^\circ + 120^\circ = 216.490\ldots^\circ \quad \text{(in range)}$$
$$216.490\ldots^\circ + 120^\circ = 336.490\ldots^\circ \quad \text{(in range)}$$

$\theta = \{23.5^\circ, 96.5^\circ, 144^\circ, 216^\circ, 264^\circ, 336^\circ\}$ to 3sf.

35 a)



$\sin\theta = \frac{AC}{k} \Rightarrow AC = k\sin\theta$.

$\cos\theta = \frac{OC}{k} \Rightarrow OC = k\cos\theta$.

$\angle OAC = 90^\circ - \theta$.

$\angle BAD = 2\theta - 90^\circ - (90^\circ - \theta) = 3\theta - 180^\circ$.

$\sin(3\theta - 180^\circ) = \frac{BD}{k} \Rightarrow BD = k\sin(3\theta - 180^\circ)$.

$\cos(3\theta - 180^\circ) = \frac{AD}{k} \Rightarrow AD = k\cos(3\theta - 180^\circ)$.

x-coordinate of B is $OC + AD$

$$= k\cos\theta + k\cos(3\theta - 180^\circ)$$
$$\equiv k\cos\theta + k(\cos 3\theta\cos 180^\circ + \sin 3\theta\sin 180^\circ)$$
$$\equiv k\cos\theta - k\cos 3\theta$$
$$\equiv k\cos\theta - k\cos(2\theta + \theta)$$
$$\equiv k\cos\theta - k(\cos 2\theta\cos\theta - \sin 2\theta\sin\theta)$$
$$\equiv k\cos\theta - k((1 - 2\sin^2\theta)\cos\theta - 2\sin\theta\cos\theta \times \sin\theta)$$
$$\equiv k\cos\theta - k(\cos\theta - 2\sin^2\theta\cos\theta - 2\sin^2\theta\cos\theta)$$
$$\equiv k\cos\theta - k(\cos\theta - 4\sin^2\theta\cos\theta)$$
$$\equiv k\cos\theta - k\cos\theta + 4k\sin^2\theta\cos\theta$$
$$\equiv 4k\sin^2\theta\cos\theta$$

y-coordinate of B is $AC + BD$

$$= k\sin\theta + k\sin(3\theta - 180^\circ)$$
$$\equiv k\sin\theta + k(\sin 3\theta\cos 180^\circ - \cos 3\theta\sin 180^\circ)$$

$$\equiv k\sin\theta - k\sin 3\theta$$
$$\equiv k\sin\theta - k\sin(2\theta+\theta)$$
$$\equiv k\sin\theta - k(\sin 2\theta\cos\theta + \cos 2\theta\sin\theta)$$
$$\equiv k\sin\theta - k(2\sin\theta\cos\theta\times\cos\theta + (2\cos^2\theta - 1)\sin\theta)$$
$$\equiv k\sin\theta - k(2\sin\theta\cos^2\theta + 2\cos^2\theta\sin\theta - \sin\theta)$$
$$\equiv k\sin\theta - k(4\sin\theta\cos^2\theta - \sin\theta)$$
$$\equiv k\sin\theta - 4k\sin\theta\cos^2\theta + k\sin\theta$$
$$\equiv 2k\sin\theta - 4k\sin\theta\cos^2\theta$$

$\therefore$ the coordinates of B are $(4k\sin^2\theta\cos\theta, 2k\sin\theta - 4k\sin\theta\cos^2\theta)$.

35 b)

$(4k\sin^2 60^\circ\cos 60^\circ, 2k\sin 60^\circ - 4k\sin 60^\circ\cos^2 60^\circ)$

$$= \left(4k\left(\frac{\sqrt{3}}{2}\right)^2\left(\frac{1}{2}\right), 2k\left(\frac{\sqrt{3}}{2}\right) - 4k\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)^2\right)$$
$$= \left(\frac{3}{2}k, \sqrt{3}k - \frac{\sqrt{3}}{2}k\right) = \left(\frac{3}{2}k, \frac{\sqrt{3}}{2}k\right)$$
$$\text{Distance} = \sqrt{\left(\frac{3}{2}k\right)^2 + \left(\frac{\sqrt{3}}{2}k\right)^2} = \sqrt{3k^2} = \sqrt{3}k$$

36)

LHS: $\sin 4x \equiv \sin(2x+2x)$
$$\equiv \sin 2x\cos 2x + \cos 2x\sin 2x$$
$$\equiv (2\sin x\cos x)(2\cos^2 x - 1) + (1 - 2\sin^2 x)(2\sin x\cos x)$$
$$\equiv 4\sin x\cos^3 x - 2\sin x\cos x + 2\sin x\cos x - 4\sin^3 x\cos x$$
$$\equiv 4\sin x\cos^3 x - 4\sin^3 x\cos x : \text{RHS}$$

37)

LHS: $\cos 4x \equiv \cos(2x+2x)$
$$\equiv \cos 2x\cos 2x - \sin 2x\sin 2x$$
$$\equiv (2\cos^2 x - 1)(2\cos^2 x - 1) - (2\sin x\cos x)(2\sin x\cos x)$$
$$\equiv 4\cos^4 x - 4\cos^2 x + 1 - 4\sin^2 x\cos^2 x$$
$$\equiv 4\cos^4 x - 4\cos^2 x + 1 - 4(1 - \cos^2 x)\cos^2 x$$
$$\equiv 4\cos^4 x - 4\cos^2 x + 1 - 4\cos^2 x + 4\cos^4 x$$
$$\equiv 8\cos^4 x - 8\cos^2 x + 1 : \text{RHS}$$

38)

LHS: $\frac{\sin 4x}{\cos x} + \frac{\cos 4x}{\sin x}$

$$\equiv \frac{4\sin x\cos^3 x - 4\sin^3 x\cos x}{\cos x} + \frac{8\cos^4 x - 8\cos^2 x + 1}{\sin x}$$

$$\equiv \frac{4\sin^2 x\cos^3 x - 4\sin^4 x\cos x}{\sin x\cos x} + \frac{8\cos^4 x - 8\cos^2 x + 1}{\sin x}$$

$$\equiv \frac{4\sin^2 x\cos^2 x - 4\sin^4 x + 8\cos^4 x - 8\cos^2 x + 1}{\sin x}$$

$$\equiv (4\sin^2 x\cos^2 x - 4\sin^4 x + 8\cos^4 x - 8\cos^2 x + 1) \times \frac{1}{\sin x}$$

$$\equiv (4\sin^2 x(1 - \sin^2 x) - 4\sin^4 x + 8(1 - \sin^2 x)^2 - 8(1 - \sin^2 x) + 1)\operatorname{cosec} x$$

$$\equiv (4\sin^2 x - 4\sin^4 x - 4\sin^4 x + 8(1 - 2\sin^2 x + \sin^4 x) - 8(1 - \sin^2 x) + 1)\operatorname{cosec} x$$

$$\equiv (4\sin^2 x - 4\sin^4 x - 4\sin^4 x + 8 - 16\sin^2 x + 8\sin^4 x - 8 + 8\sin^2 x + 1)\operatorname{cosec} x$$

$$\equiv (1 - 4\sin^2 x)\operatorname{cosec} x$$

$$\equiv \left(1 - 4\left(\frac{1 - \cos 2x}{2}\right)\right)\operatorname{cosec} x$$

$$\equiv (1 - 2(1 - \cos 2x))\operatorname{cosec} x$$

$$\equiv (1 - 2 + 2\cos 2x)\operatorname{cosec} x$$

$$\equiv (2\cos 2x - 1)\operatorname{cosec} x : \text{RHS}$$